

Attack–Fault Ambiguity Against a Strategic Attacker: The Stealth–Harm Frontier, Exploitable Commitments, and the Verification Margin

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Abstract—Analyses of attack–fault ambiguity at the electronic trading gateway have so far treated the attacker as fixed: its injection is a given alternative to a benign fault, and the operator’s decision rule is designed against it. An attacker who knows the rule can instead choose an injection that imitates the fault. We formulate this as a leader–follower game in a linear–Gaussian order–flow model. The operator commits to a containment rule, possibly with a costly verification step; the attacker, knowing the rule, then chooses a payload whose observation law lies within total-variation distance ε of the fault’s. We ask four questions. How much excess harm, beyond what the imitated fault causes, can an ε -stealthy attacker extract, as a function of ε ? What remains of it as ε tends to zero, under output-level and under state-level injection? Can an attacker exploit a threshold rule committed to a nominal attack model, and what should the operator commit to instead? And what detection margin does a verification step that reads a channel order flow cannot disguise guarantee against such an attacker?

I. INTRODUCTION

An incident at an electronic trading gateway, such as an algorithm that suddenly sends abnormal order flow, can have two causes: a *compromise*, in which an adversary controls the order flow, or a benign *fault*, such as a malfunctioning algorithm. The operator must decide whether to contain the source, at the cost of disrupting a legitimate client if the cause is a fault, or to leave it, at the cost of the harm a compromise goes on to do. Throughout, *the framework* means the setting of Papers 1 and 2 of this series: an operator observes the order flow over a window, forms the posterior probability that the cause is a compromise, and chooses between containing and leaving, possibly after a costly verification. A *rule* is the operator’s committed decision rule in this framework: a map from what it observes (the order flow and, when it verifies, the outcome of the check) to the probability of containing (Definition 5).

Paper 1 [1], a working paper, models the order flow as a linear–Gaussian system and characterizes when a compromise and a fault cannot be told apart from order flow alone, and what that ambiguity costs the operator. Paper 2 [2], also a working paper, adds verification as a

costed action that reads the authorization channel rather than the order flow, and derives a leave / verify / contain rule with two belief thresholds; that derivation has not yet been independently verified. Both treat the attacker as fixed: its injection is a given alternative to the fault, and the rule is designed against it. Paper 3 [3], in preparation, is to make the response sequential with graded containment; this paper uses none of its results.

This paper asks whether the framework survives an adversary who knows the rule. We model the interaction as a leader–follower game, as in Stackelberg security games [4]: the operator commits to a rule, and the attacker, knowing it, chooses its injection. Commitment is the natural reading at a trading gateway, where pre-trade controls and kill-switch procedures are documented, tested and audited before any incident; and a rule that is safe only while it is secret offers no security guarantee. The attacker is ε -*stealthy*: the law of the order flow under its injection lies within total-variation distance ε of the law under the fault it imitates.

We address four questions. First, how much excess harm, beyond what the imitated fault causes, can an ε -stealthy attacker with a given payload budget extract, and how does this stealth–harm frontier depend on ε (Section IV)? Second, what remains of the frontier as ε tends to zero, under output-level and under state-level injection? Third, can an attacker exploit a threshold rule committed to a nominal attack model, and what should the operator commit to instead? Fourth, what detection margin does a verification step, which reads a channel order flow cannot disguise, guarantee against a strategic attacker, and where should the operator verify?

Section II positions the paper. Section III states the game, the attacker’s information and the payoffs. Section IV treats the four questions, Section V illustrates them numerically, Section VI discusses simultaneous play and the limits of the model, and Section VII concludes.

II. RELATED WORK

a) *Stealthy attacks on control systems*: Attacks that an output-based monitor cannot detect correspond to the zero dynamics of a linear system and, in the attack-free case, to its unobservable subspace [5]. Paper 1 [1] uses this subspace to characterize when a compromise and a fault remain confounded under state-level injection; this paper takes the mechanism as given and studies its consequence

for a strategic attacker. Stealth with a tolerance was introduced as an ε -bound on the Kullback–Leibler divergence between attacked and nominal measurement sequences, with the largest performance degradation given as a function of ε [6]. Optimal strictly stealthy and ε -stealthy sensor attacks on a linear-quadratic-Gaussian cost have since been derived, with an upper bound on the attack performance as a function of the stealth level [7]. Analytic worst-case Kullback–Leibler ε -stealthy attacks on remote state estimation correct earlier suboptimal solutions [8], and worst-case stealthy attacks have also been characterized for stochastic event-based state estimation [9]. A per-step Kullback–Leibler stealth constraint leads to an online attack computed by a convex program, with a unique optimal attack at zero threshold [10]. A parameterized notion of ε -stealthiness lets an attack on remote state estimation pass the detector at a lower alarm rate [11], and an ε -stealthy attacker that also uses intercepted data and side information maximizes the estimation error covariance analytically [12]. Here the stealth budget is a total-variation distance, which bounds the detection gap of any rule directly, and the reference law is that of a benign fault rather than of the attack-free system.

b) Detection, harm and defence design against stealthy attacks: A detector limits the set of biases a stealthy attacker can exert, which bounds how long a system can remain under stealthy attack before its safety is violated [13]. On the defender’s side, the expected worst-case impact of stealthy data injection has been used to allocate monitors in a Stackelberg framework [14], and optimal sensor allocations that minimize the expected impact of stealthy injection attacks have been computed for networks [15]. In a moving-target defence, the defender randomizes its detector threshold in a Bayesian game with a prior over the attacker’s goals, and its cost adds the cost of false alarms to the cost the attack induces [16]. In this paper the operator commits to a posterior containment rule rather than allocating sensors or randomizing a threshold, the attacker imitates a benign fault, and the limit of the excess harm as ε tends to zero is tied to the pair (A, C) .

c) Adversarial classification and hypothesis testing: Adversarial classification casts a classifier and an adversary who modifies its inputs as players of a game [17]. Closest to this paper, game-theoretic Neyman–Pearson detection designs an evasion-aware detector against a strategic evader and summarizes what each side can achieve by an equilibrium receiver operating characteristic [18]. Over controlled continuous-time dynamics, a zero-sum game between a covert attacker and a detector has bias injection and a likelihood-ratio test as its unique saddle point, with type-II error exponents in the data length [19]. Here the game is leader–follower rather than zero-sum, the operator minimizes an expected cost with a posterior containment rule instead of a Neyman–Pearson criterion, the benign alternative is a fault, the observations come from a linear dynamical system over a finite window, and a verification channel is available.

d) Attacks, faults and active verification: A designed watermark combined with adaptive estimation discriminates sensor replay attacks from sensor bias faults, and conditions characterize which attacks and faults can be told apart [20]. Watermarking raises the Kullback–Leibler divergence between the attacked and nominal joint statistics at a cost in control performance, and the optimal watermark variance for a fixed cost increase has been derived [21]. When to watermark can itself be posed as a stochastic control problem, whose optimal policy consists of two thresholds on the posterior probability of attack [22]; the two-threshold shape of a verification rule is therefore not new. In these works the attacker does not respond to the check; here the attacker knows the committed two-threshold rule with verification of Paper 2 [2] and best-responds to it, and the question is which detection margin remains.

e) Security games: In Stackelberg security games a defender commits to a strategy that the attacker observes before responding, and the relation between Stackelberg and Nash strategies in such games has been studied in detail [4]. Recent instances include a defender who commits to a resource allocation against a damage-maximizing attacker whose attacks spread over a network [23], and a defender who commits to a mixed sequential strategy against multi-drone attacks, computed as an approximate strong Stackelberg equilibrium [24]. This paper adopts the commitment form with the operator as leader and returns to simultaneous play in Section VI.

f) Manipulation in electronic markets: In high-frequency trading, the optimal spoofing strategy under multilevel order-book imbalance has been used to characterize and detect spoofing [25]. There the detector is not a committed leader, and no benign fault is confounded with the manipulation.

III. MODEL

This section states the game. The operator, as leader, commits to a rule and an observation window; the attacker, as follower, observes both and chooses an injection within a payload budget; nature then draws the cause, compromise or fault (Definition 1). Observations follow the linear-Gaussian order-flow model of Paper 1 [1] on a finite window W of observation times after the incident’s onset, under output-level or state-level injection (Definition 2). The attacker’s payload is the part of its injection that differs from the fault’s (Definition 3), and it is ε -stealthy when the law of the observations on W under its injection lies within total-variation distance ε of the law under the fault (Definition 4). The operator chooses among rules without verification, including threshold rules committed to a nominal attack model, and rules with a verification step of the form proposed in Paper 2 [2] (Definition 5). The attacker’s *excess harm* is a linear functional of its payload, the harm it extracts beyond what the imitated fault would cause (Definition 7); the stealth–harm frontier $\mathcal{F}_W(\varepsilon)$ is the largest excess harm over ε -stealthy payloads, and the harm ceiling is its limit as ε tends to zero (Definition 9).

A. The game

Definition 1 (Players, timing, game form). There are two players, the *operator* (leader) and the *attacker* (follower), and a move of nature. (1) The operator commits to a rule π from a class Π (Definition 5) and to an observation window W . (2) The attacker observes π and W and chooses an injection vector x_K from its admissible set (Definition 3). (3) Nature draws the cause $M = K$ with probability $p \in (0, 1)$ and $M = F$ otherwise; under F the fault instance x_F is fixed (Assumption 4), under K the injection is x_K . (4) The operator observes Y_W (and, under a rule with verification, the verification outcome Z) and applies π . The attacker's choice is a pure strategy; the operator's rule may be randomized (values in $[0, 1]$). The operator does not revise π after observing Y_W beyond what π itself prescribes (commitment).

Definition 2 (Observation model and stealth geometry). The system follows Paper 1's output-level model (Paper 1, Definition D1) or its state-level model (Paper 1, Definition D2), with a finite window $W \supseteq \{\tau, \dots, \tau + H - 1\}$, $H \geq 1$. For an injection vector $x \in \mathbb{R}^n$, write P_x^W for the law of Y_W when the injection from the onset on is x . Under Assumption 1 every P_x^W is Gaussian with a covariance $V_W \succ 0$ that does not depend on x , and mean $\mu_W + G_W x$, where G_W stacks, for $t \in W$, the zero block if $t < \tau$, and for $t = \tau + j$, $j \geq 0$,

$$C \text{ (output-level),} \quad C \sum_{i=0}^j A^i \text{ (state-level).}$$

The *stealth matrix* is $M_W = G_W^\top V_W^{-1} G_W \succeq 0$, and the *blind subspace* is $\mathcal{B}_W = \ker G_W = \ker M_W$.

Definition 3 (Attacker strategy, payload, budget). The attacker chooses a severity $\theta_K \in \mathbb{R}$ and a direction $b_K \in \mathbb{R}^n$; only the injection vector $x_K = \theta_K b_K$ enters the observation law (Definition 2), so a strategy is identified with x_K . The *payload* is $\Delta = x_K - x_F$, the part of the injection that differs from the fault's (Paper 1's Δ_0). The admissible set is the payload budget

$$\mathcal{X}_r = \{x_K \in \mathbb{R}^n : \|x_K - x_F\|_2 \leq r\}, \quad r > 0.$$

Definition 4 (ε -stealthy). For $\varepsilon \in [0, 1]$, an attacker strategy x_K is ε -stealthy on W if $\text{TV}(P_{x_K}^W, P_{x_F}^W) \leq \varepsilon$ (total variation as in Paper 1, Definition D10). The ε -stealthy set is $\mathcal{E}_W(\varepsilon) = \{x_K \in \mathcal{X}_r : \text{TV}(P_{x_K}^W, P_{x_F}^W) \leq \varepsilon\}$. Under Assumption 1, with $d_W(\Delta) = \sqrt{\Delta^\top M_W \Delta}$,

$$\text{TV}(P_{x_K}^W, P_{x_F}^W) = 2\Phi(d_W(\Delta)/2) - 1, \quad \mathcal{E}_W(\varepsilon) = \{x_K \in \mathcal{X}_r : d_W(x_K - x_F) \leq \rho_\varepsilon\},$$

with $\rho_1 = +\infty$. $\varepsilon = 0$ is exact confounding on W (Paper 1, Definition D4).

Definition 5 (Operator rules). A *rule* on W is a measurable map π from observations to the probability of containing (Paper 1, Definition D9), possibly after a verification step. The classes used are: (i) *Committed threshold rule* $\pi_{\bar{x}}^{\text{thr}}$. The operator fixes a *nominal compromise model* $\bar{x} \neq x_F$ (the compromise instance it plans against) and contains iff the nominal posterior

$\bar{q}(Y_W) = p f_{\bar{x}}^W(Y_W) / (p f_{\bar{x}}^W(Y_W) + (1-p) f_{x_F}^W(Y_W))$ satisfies $\bar{q}(Y_W) \geq q^* = b/(a+b)$ (Paper 2, Lemma 1). (ii) *Committed two-threshold rule with verification* $\pi_{\bar{x}}^{\text{ver}}$. With the same $\bar{q}$ and Paper 2's thresholds $q_1 \leq q_2$ computed from $(a, b, s_v, t_v, c_v, r_K, \tau_v)$ (Paper 2, Theorem 1): leave if $\bar{q} \leq q_1$; if $q_1 < \bar{q} < q_2$, verify, then contain on $Z = +$ and leave on $Z = -$; contain if $\bar{q} \geq q_2$. (iii) *General rule*: any π (for the operator's optimal commitment); rules with verification are written as the triple of the verify probability δ and the containment probabilities after $Z = \pm$ and without verification, as in Paper 2, Definition 4. (iv) *Throttle* (Paper 3): a later extension, not defined here.

Definition 6 (Detection, false alarm, verification mass). For a rule π let $\phi_\pi^M(y) \in [0, 1]$ be the probability of containing given $Y_W = y$ and cause M (for a rule without verification $\phi_\pi^K = \phi_\pi^F = \pi$; for $\pi_{\bar{x}}^{\text{ver}}$, $\phi^K = \mathbb{I}[\bar{q} \geq q_2] + s_v \mathbb{I}[q_1 < \bar{q} < q_2]$ and $\phi^F = \mathbb{I}[\bar{q} \geq q_2] + (1-t_v) \mathbb{I}[q_1 < \bar{q} < q_2]$). The *detection probability* of strategy x_K is $\beta_\pi(x_K) = \mathbb{E}_{x_K}[\phi_\pi^K(Y_W)]$, the *false-alarm probability* is $\alpha_\pi = \mathbb{E}_{x_F}[\phi_\pi^F(Y_W)]$, and for a rule with verification the *fault verification mass* is $m_F = P_{x_F}(q_1 < \bar{q}(Y_W) < q_2)$ (Paper 2's verification mass, Definition 5, evaluated on the fault law).

B. Harm, payoffs and the stealth-harm frontier

Definition 7 (Harm and payoffs). Fix a *harm vector* $g \in \mathbb{R}^n$. The attacker's *excess harm* is the linear functional $h(\Delta) = g^\top \Delta$ of its payload: the harm it extracts beyond what the fault it imitates would cause. The attacker's payoff against rule π is the expected excess harm extracted before containment, net of a capture loss $c_A \geq 0$:

$$U_A(\pi, x_K) = h(x_K - x_F) (1 - \beta_\pi(x_K)) - c_A \beta_\pi(x_K).$$

The operator's expected cost is Paper 1's loss table (Definition D8, general symbols) with the leave-a-compromise entry raised by the extracted harm, $H_K + h(\Delta)$:

$$J(\pi, x_K) = p \left(\beta_\pi(x_K) C_K + (1 - \beta_\pi(x_K)) (H_K + h(\Delta)) \right) + (1-p) (\alpha_\pi C_F - \text{Ver}_\pi)$$

where Ver_π is the expected verification cost $c_v + \mathbb{I}[M = K] r_K \tau_v$ incurred on the event that π verifies (Paper 2, Definition 4), and zero for a rule without verification. The game is not zero-sum ($U_A \neq -J$ in general).

Definition 8 (Strong Stackelberg equilibrium). For a rule π , the attacker's ε -stealthy best-response set is $\text{BR}_\varepsilon(\pi) = \arg \max_{x \in \mathcal{E}_W(\varepsilon)} U_A(\pi, x)$ (non-empty under Assumption 3 when $U_A(\pi, \cdot)$ is upper semicontinuous). A pair (π^*, x^*) is a *strong Stackelberg equilibrium* in class Π at stealth level ε if $x^* \in \text{BR}_\varepsilon(\pi^*)$, ties being broken in the operator's favour (x^* minimizes $J(\pi^*, \cdot)$ over $\text{BR}_\varepsilon(\pi^*)$), and π^* minimizes $\min_{x \in \text{BR}_\varepsilon(\pi)} J(\pi, x)$ over $\pi \in \Pi$ (an infimum if not attained). Under simultaneous play, the corresponding notion is a Nash equilibrium of the same payoffs, the attacker possibly randomizing over $\mathcal{E}_W(\varepsilon)$.

Definition 9 (Stealth-harm frontier, detection-harm frontier, harm ceiling). The *stealth-harm frontier* on W is

$$\mathcal{F}_W(\varepsilon) = \sup\{h(x - x_F) : x \in \mathcal{E}_W(\varepsilon)\}, \quad \varepsilon \in [0, 1],$$

which does not depend on the operator's rule. The *detection-harm frontier* against a rule π is $\Psi_\pi(\beta) = \sup\{h(x - x_F) : x \in \mathcal{X}_r, \beta_\pi(x) \leq \beta\}$, $\beta \in [0, 1]$. The *harm ceiling* is $\bar{\mathcal{F}}_W = \lim_{\varepsilon \downarrow 0} \mathcal{F}_W(\varepsilon)$ (the limit exists since $\mathcal{F}_W$ is nondecreasing).

C. Standing assumptions

Assumption 1 (Paper 1's linear-Gaussian model). Paper 1's standing assumptions (A1)–(A6) hold, with a finite window as in Definition 2; hence $V_W \succ 0$. (A, C, Q, Σ) , the onset τ and the window W are common to both causes and known to both players.

Assumption 2 (Information). The attacker knows the committed rule π as a map from observations (and verification outcomes) to actions, the window W , the model of Assumption 1 and the fault instance x_F . It need not know p , the loss table or the nominal model $\bar{x}$ separately: its best response depends on them only through π . The operator knows p , the loss table, the model and x_F , but not the attacker's choice x_K , which enters its rule only through the nominal model $\bar{x}$ it commits to.

Assumption 3 (Payload budget). The attacker's admissible set is $\mathcal{X}_r$ (Definition 3) for a known $r > 0$. (Compact and convex; any compact convex admissible set containing x_F could replace it at the cost of the closed forms.)

Assumption 4 (One fixed fault instance). Under F the injection is a fixed, known $x_F = \theta_F b_F$ (Paper 1's pairwise-instance reading, Definition D3); no distribution over fault instances is modelled.

Assumption 5 (Cost orderings). $a = H_K - C_K > 0$ and $b = C_F - H_F > 0$ (Paper 2, Assumption 1); for rules with verification, Paper 2's Assumptions 3 and 4 hold with sensitivity s_v , specificity t_v , cost c_v , delay τ_v and delay harm rate r_K .

Assumption 6 (Verification is not manipulable by order flow). Paper 2's separate-channel assumption (Assumption 2) holds for every attacker strategy: given the cause, Z is independent of Y_W , and (s_v, t_v) do not depend on x_K . (An attacker who also attacks the authorization channel, for instance with stolen credentials, violates this assumption; that case is out of scope.)

IV. RESULTS

This section states the results on the three questions of the title: the stealth-harm frontier and the harm ceiling (Section IV-A), the exploitability of a committed threshold rule (Section IV-B), and the verification margin (Section IV-C). Each result has a written proof, given after its statement; the proofs have been checked by the author and have not yet been independently verified.

A. The stealth-harm frontier and the harm ceiling

Proposition 1 bounds, for every committed rule, the detection probability of an attacker whose observation law is within total variation ε of the fault's.

Proposition 1 (Total-variation detection cap). *Assume Assumption 1. Let $\varepsilon \in [0, 1]$ and let $x_K \in \mathbb{R}^n$ satisfy $\text{TV}(P_{x_K}^W, P_{x_F}^W) \leq \varepsilon$ (in particular, any $x_K \in \mathcal{E}_W(\varepsilon)$). (a) For every measurable ϕ from observations on W to $[0, 1]$, $|\mathbb{E}_{x_K}[\phi(Y_W)] - \mathbb{E}_{x_F}[\phi(Y_W)]| \leq \varepsilon$. (b) For every rule π without verification, $\alpha_\pi - \varepsilon \leq \beta_\pi(x_K) \leq \alpha_\pi + \varepsilon$. (c) Assume in addition Assumption 6. For every rule with verification, written as in Definition 5(iii) as $(\delta, \pi_0, \pi_+, \pi_-)$, let $\phi^K = (1 - \delta)\pi_0 + \delta(s_v\pi_+ + (1 - s_v)\pi_-)$ and $\phi^F = (1 - \delta)\pi_0 + \delta((1 - t_v)\pi_+ + t_v\pi_-)$ (for $\pi_{\bar{x}}^{\text{ver}}$ these are the maps of Definition 6). Then $\beta_\pi(x_K) = \mathbb{E}_{x_K}[\phi^K(Y_W)]$, $\alpha_\pi = \mathbb{E}_{x_F}[\phi^F(Y_W)]$, and $\mathbb{E}_{x_F}[\phi^K(Y_W)] - \varepsilon \leq \beta_\pi(x_K) \leq \mathbb{E}_{x_F}[\phi^K(Y_W)] + \varepsilon$. (d) (Tightness.) For every $x \in \mathbb{R}^n$, $\sup_\pi (\beta_\pi(x) - \alpha_\pi) = \text{TV}(P_x^W, P_{x_F}^W)$, the supremum over rules without verification, attained by the rule that contains iff $f_x^W(Y_W) > f_{x_F}^W(Y_W)$.*

Proof. Write $P = P_{x_K}^W$, $P' = P_{x_F}^W$, and let f, f' be their densities with respect to $\mu = P + P'$ (under Assumption 1 both are Gaussian with $V_W \succ 0$, so Lebesgue densities also serve). Let $A_+ = \{f > f'\}$.

(a). For measurable ϕ with values in $[0, 1]$,

$$\mathbb{E}_P[\phi] - \mathbb{E}_{P'}[\phi] = \int \phi(f - f') d\mu \leq \int_{A_+} (f - f') d\mu = P(A_+) - P'(A_+) \leq \varepsilon$$

since $\phi \leq 1$ on A_+ and $\phi \geq 0$ off it. Exchanging P and P' (TV is symmetric) gives the other sign. Hence $|\mathbb{E}_P[\phi] - \mathbb{E}_{P'}[\phi]| \leq \text{TV}(P, P') \leq \varepsilon$.

(b). For a rule without verification, $\phi_\pi^K = \phi_\pi^F = \pi$ (Definition 6), a measurable map into $[0, 1]$, so $\beta_\pi(x_K) = \mathbb{E}_P[\pi]$ and $\alpha_\pi = \mathbb{E}_{P'}[\pi]$. Apply (a) with $\phi = \pi$.

(c). Given $Y_W = y$ and cause M , the operator verifies with probability $\delta(y)$, and then contains with probability $\pi_Z(y)$; otherwise it contains with probability $\pi_0(y)$. By Assumption 6, Z is independent of Y_W given the cause, with $P(Z = + | K) = s_v$ and $P(Z = - | F) = t_v$, and these do not depend on x_K . Hence the containment probability given $Y_W = y$ is $\phi^K(y)$ under K and $\phi^F(y)$ under F , as defined in the statement. Both are convex combinations of values in $[0, 1]$, so they are measurable maps into $[0, 1]$. Averaging over Y_W under the law of each cause gives $\beta_\pi(x_K) = \mathbb{E}_P[\phi^K]$ and $\alpha_\pi = \mathbb{E}_{P'}[\phi^F]$. For $\pi_{\bar{x}}^{\text{ver}}$ ($\delta = \mathbb{I}[q_1 < \bar{q} < q_2]$, $\pi_0 = \mathbb{I}[\bar{q} \geq q_2]$, $\pi_+ = 1$, $\pi_- = 0$) these are the maps of Definition 6. Apply (a) with $\phi = \phi^K$: $|\mathbb{E}_P[\phi^K] - \mathbb{E}_{P'}[\phi^K]| \leq \varepsilon$.

(d). By (b) with $\varepsilon = \text{TV}(P_x^W, P_{x_F}^W)$, every rule without verification has $\beta_\pi(x) - \alpha_\pi \leq \text{TV}(P_x^W, P_{x_F}^W)$. The rule $\pi = \mathbb{I}_{A_+}$ (with $P = P_x^W$) gives $\beta_\pi(x) - \alpha_\pi = P(A_+) - P'(A_+) = \int_{A_+} (f - f') d\mu = \frac{1}{2} \int |f - f'| d\mu = \text{TV}(P, P')$, where the middle equality uses $\int (f - f') d\mu = 0$. So the supremum is attained. $\square$

Proposition 2 gives the shape of the stealth-harm frontier $\mathcal{F}_W(\varepsilon)$ of Definition 9 as a function of ε .

Proposition 2 (Shape of the stealth-harm frontier). *Assume Assumption 1 and Assumption 3, write $M = M_W$ and, for $\rho \in [0, \infty)$,*

$$f(\rho) = \max\{g^\top \Delta : \|\Delta\|_2 \leq r, \Delta^\top M \Delta \leq \rho^2\},$$

so that $\mathcal{F}_W(\varepsilon) = f(\rho_\varepsilon)$ for $\varepsilon \in [0, 1)$ (the maximum is attained). Let $\rho_{\text{sat}} = r\sqrt{g^\top M g}/\|g\|_2$ ($\rho_{\text{sat}} = 0$ if $g = 0$) and $\varepsilon_{\text{sat}} = 2\Phi(\rho_{\text{sat}}/2) - 1 \in [0, 1)$. (a) f is nondecreasing, concave and continuous on $[0, \infty)$. $\mathcal{F}_W$ is nondecreasing and continuous on $[0, 1]$, with $\mathcal{F}_W(1) = r\|g\|_2$. (b) (Saturation.) $\mathcal{F}_W(\varepsilon) = r\|g\|_2$ if $\varepsilon \geq \varepsilon_{\text{sat}}$ and $\mathcal{F}_W(\varepsilon) < r\|g\|_2$ if $\varepsilon < \varepsilon_{\text{sat}}$. Moreover $\varepsilon_{\text{sat}} = 0$ if and only if $g \in \ker G_W$. (c) (Without the budget.) For $\rho \geq 0$, $\sup\{g^\top \Delta : \Delta^\top M \Delta \leq \rho^2\} = \rho\sqrt{g^\top M^+ g}$ if $g \in \text{range}(M)$, and $+\infty$ otherwise; for $g \in \text{range}(M) \setminus \{0\}$ and $\rho > 0$ it is attained at $\rho M^+ g / \sqrt{g^\top M^+ g}$. (d) (With the budget: a one-dimensional search.) Let $g \neq 0$. Put $\rho_b = r\sqrt{g^\top M^+ g} / \|M^+ g\|_2$ if $g \in \text{range}(M)$ and $\rho_b = 0$ otherwise; for $\nu \geq 0$ put $\Delta(\nu) = (I + \nu M)^{-1} g$ and $R(\nu) = \Delta(\nu)^\top M \Delta(\nu) / \|\Delta(\nu)\|_2^2$. Then R is continuous and non-increasing on $[0, \infty)$, $R(0) = (\rho_{\text{sat}}/r)^2$, $\lim_{\nu \rightarrow \infty} R(\nu) = (\rho_b/r)^2$, so $\rho_b \leq \rho_{\text{sat}}$, and R is strictly decreasing if $\rho_b < \rho_{\text{sat}}$. Further: (d1) if $\rho \geq \rho_{\text{sat}}$, $f(\rho) = r\|g\|_2$, attained at $rg/\|g\|_2$; (d2) if $g \in \text{range}(M)$ and $\rho \leq \rho_b$, $f(\rho) = \rho\sqrt{g^\top M^+ g}$, attained as in (c) (budget inactive); (d3) if $\rho_b < \rho < \rho_{\text{sat}}$, the equation $R(\nu) = (\rho/r)^2$ has a unique root $\nu_\rho \in (0, \infty)$, and $f(\rho) = r g^\top \Delta(\nu_\rho) / \|\Delta(\nu_\rho)\|_2$, attained at $r \Delta(\nu_\rho) / \|\Delta(\nu_\rho)\|_2$ (both constraints active). Together with $f(0) = r\|\Pi_{\ker G_W} g\|_2$ (Theorem 1(a)), the cases (d1)–(d3) cover every $\rho \geq 0$.

Proof. Write $K(\rho) = \{\Delta : \|\Delta\|_2 \leq r, \Delta^\top M \Delta \leq \rho^2\}$, $\|\cdot\| = \|\cdot\|_2$ and $\|\Delta\|_M = \sqrt{\Delta^\top M \Delta}$, a seminorm since $M \succeq 0$.

Step 0 (the frontier is $f(\rho_\varepsilon)$). By Definition 4, for $\varepsilon \in [0, 1)$, $x \in \mathcal{E}_W(\varepsilon)$ iff $\Delta = x - x_F \in K(\rho_\varepsilon)$, and $h(x - x_F) = g^\top \Delta$. $K(\rho)$ is compact (closed, inside the ball) and non-empty ($0 \in K(\rho)$), so the maximum of the continuous map $\Delta \mapsto g^\top \Delta$ is attained and $\mathcal{F}_W(\varepsilon) = f(\rho_\varepsilon)$. For $\varepsilon = 1$, $\mathcal{E}_W(1) = \mathcal{X}_r$ and $\mathcal{F}_W(1) = \max_{\|\Delta\|_2 \leq r} g^\top \Delta = r\|g\|$ (Cauchy–Schwarz, attained at $rg/\|g\|$ if $g \neq 0$).

(a). Monotonicity: $K(\rho) \subseteq K(\rho')$ for $\rho \leq \rho'$. Concavity: let $\Delta_i \in K(\rho_i)$ attain $f(\rho_i)$, $i = 1, 2$, and $\lambda \in [0, 1]$. Then $\Delta = \lambda \Delta_1 + (1 - \lambda) \Delta_2$ has $\|\Delta\| \leq r$ and $\|\Delta\|_M \leq \lambda \rho_1 + (1 - \lambda) \rho_2$ (triangle inequality for the seminorm), so $\Delta \in K(\lambda \rho_1 + (1 - \lambda) \rho_2)$ and $f(\lambda \rho_1 + (1 - \lambda) \rho_2) \geq g^\top \Delta = \lambda f(\rho_1) + (1 - \lambda) f(\rho_2)$. A finite concave function on $[0, \infty)$ is continuous on $(0, \infty)$. Right-continuity at 0: let $\rho_k \downarrow 0$ and $\Delta_k \in K(\rho_k)$ attain $f(\rho_k)$. All Δ_k lie in the compact ball; take a subsequence along which $\Delta_k \rightarrow \bar{\Delta}$ and $f(\rho_k) \rightarrow \limsup_k f(\rho_k)$. Then $\|\bar{\Delta}\| \leq r$ and $\bar{\Delta}^\top M \bar{\Delta} = \lim \Delta_k^\top M \Delta_k \leq \lim \rho_k^2 = 0$, so $\bar{\Delta} \in K(0)$ and $\limsup_k f(\rho_k) = g^\top \bar{\Delta} \leq f(0)$. With monotonicity, $f(\rho_k) \rightarrow f(0)$. The map $\varepsilon \mapsto \rho_\varepsilon = 2\Phi^{-1}((1 + \varepsilon)/2)$ is continuous and strictly increasing on $[0, 1)$, with $\rho_0 = 0$ and $\rho_\varepsilon \rightarrow \infty$ as $\varepsilon \uparrow 1$. So $\mathcal{F}_W = f \circ \rho_\varepsilon$ is nondecreasing and continuous on $[0, 1)$. By (d1) below (or directly:

$rg/\|g\| \in K(\rho)$ for $\rho \geq \rho_{\text{sat}}$), $\mathcal{F}_W(\varepsilon) = r\|g\| = \mathcal{F}_W(1)$ for all ε close enough to 1, which gives continuity at 1.

(b). If $g = 0$, $f \equiv 0 = r\|g\|$, $\rho_{\text{sat}} = 0$, $\varepsilon_{\text{sat}} = 0$, and $g \in \ker G_W$: the claims hold. Let $g \neq 0$. The linear map $\Delta \mapsto g^\top \Delta$ has the unique maximizer $\Delta^* = rg/\|g\|$ on the ball (equality case of Cauchy–Schwarz). As $K(\rho)$ is inside the ball, $f(\rho) = r\|g\|$ iff $\Delta^* \in K(\rho)$ iff $\|\Delta^*\|_M = r\sqrt{g^\top M g}/\|g\| = \rho_{\text{sat}} \leq \rho$; otherwise $f(\rho) < r\|g\|$. Since $\varepsilon \mapsto \rho_\varepsilon$ is strictly increasing with inverse $\rho \mapsto 2\Phi(\rho/2) - 1$, $\rho_\varepsilon \geq \rho_{\text{sat}}$ iff $\varepsilon \geq \varepsilon_{\text{sat}}$. Finally $\varepsilon_{\text{sat}} = 0$ iff $\rho_{\text{sat}} = 0$ iff $g^\top M g = 0$ iff $M g = 0$ (as $M \succeq 0$) iff $g \in \ker M = \ker G_W$ (Definition 2: $\Delta^\top M \Delta = \|V_W^{-1/2} G_W \Delta\|^2$ and $V_W \succ 0$).

(c). If $g \notin \text{range}(M) = (\ker M)^\perp$, then $u = \Pi_{\ker M} g \neq 0$ and $\Delta = tu$ is feasible for every $\rho \geq 0$ and $t > 0$ ($\Delta^\top M \Delta = 0$), with $g^\top \Delta = t\|u\|^2 \rightarrow \infty$. If $g = Mv$ for some v , then for feasible Δ , $g^\top \Delta = v^\top M \Delta \leq \|v\|_M \|\Delta\|_M \leq \rho \|v\|_M$ (Cauchy–Schwarz for the semi-inner product $\langle u, u' \rangle_M = u^\top M u'$), and $\|v\|_M^2 = v^\top M v = v^\top M M^+ M v = g^\top M^+ g$. For $g \neq 0$, $\rho > 0$, the point $\Delta_u = \rho M^+ g / \sqrt{g^\top M^+ g}$ (note $g^\top M^+ g = \|v\|_M^2 > 0$, since $M v = g \neq 0$) has $\Delta_u^\top M \Delta_u = \rho^2 g^\top M^+ M M^+ g / (g^\top M^+ g) = \rho^2$ and $g^\top \Delta_u = \rho \sqrt{g^\top M^+ g}$. For $\rho = 0$ or $g = 0$ the value 0 is attained at $\Delta = 0$.

(d): properties of R . Let $M = \sum_i m_i e_i e_i^\top$ be an eigen-decomposition, $m_i \geq 0$, and $\gamma_i = e_i^\top g$. Then $\Delta(\nu) = \sum_i \gamma_i (1 + \nu m_i)^{-1} e_i$, which is non-zero for $g \neq 0$, and

$$R(\nu) = \frac{\sum_i w_i(\nu) m_i}{\sum_i w_i(\nu)}, \quad w_i(\nu) = \frac{\gamma_i^2}{(1 + \nu m_i)^2},$$

a weighted mean of the m_i over the index set $I_g = \{i : \gamma_i \neq 0\}$, which does not depend on ν . R is continuous and differentiable on $[0, \infty)$. With $\ell_i(\nu) = w'_i(\nu)/w_i(\nu) = -2m_i/(1 + \nu m_i)$ and $\mathbb{E}_w$, Cov_w the mean and covariance under the normalized weights,

$$R'(\nu) = \text{Cov}_w(m, \ell) = \frac{1}{2} \sum_{i,j} \bar{w}_i \bar{w}_j (m_i - m_j)(\ell_i - \ell_j), \quad \bar{w}_i = w_i / \sum_i w_i$$

The map $m \mapsto -2m/(1 + \nu m)$ is strictly decreasing on $[0, \infty)$ (derivative $-2/(1 + \nu m)^2$), so every term is ≤ 0 , and $R' < 0$ unless m_i is constant over I_g . Thus R is nonincreasing, and strictly decreasing unless the m_i , $i \in I_g$, all coincide. $R(0) = g^\top M g / \|g\|^2 = (\rho_{\text{sat}}/r)^2$. As $\nu \rightarrow \infty$: if some $i \in I_g$ has $m_i = 0$ (i.e. $g \notin \text{range}(M)$), its weight stays γ_i^2 while the weights with $m_i > 0$ tend to 0, so $R(\nu) \rightarrow 0 = (\rho_b/r)^2$. If $g \in \text{range}(M)$, all $i \in I_g$ have $m_i > 0$ and $\nu^2 w_i(\nu) \rightarrow \gamma_i^2/m_i^2$, so $R(\nu) \rightarrow \sum_{i \in I_g} \gamma_i^2/m_i / \sum_{i \in I_g} \gamma_i^2/m_i^2 = g^\top M^+ g / \|M^+ g\|^2 = (\rho_b/r)^2$. Monotonicity gives $\rho_b \leq \rho_{\text{sat}}$. If the m_i , $i \in I_g$, all equal some m , then $R \equiv m$ and $\rho_b = \rho_{\text{sat}}$ (if $m > 0$, $g \in \text{range } M$ and both limits equal m ; if $m = 0$, both are 0). Hence $\rho_b < \rho_{\text{sat}}$ implies that the m_i are not all equal on I_g , so R is strictly decreasing.

(d1). This is (b): $\rho \geq \rho_{\text{sat}}$ iff $rg/\|g\| \in K(\rho)$.

(d2). Let $g \in \text{range}(M)$, $\rho \leq \rho_b$. If $\rho = 0$, $f(0) = 0$ (every $\Delta \in K(0)$ is in $\ker M \perp g$) $= \rho \sqrt{g^\top M^+ g}$. If $\rho > 0$, the point Δ_u of (c) has $\|\Delta_u\| = \rho \|M^+ g\| / \sqrt{g^\top M^+ g} \leq \rho_b \|M^+ g\| / \sqrt{g^\top M^+ g} = r$, so it is in $K(\rho)$, and it maximizes $g^\top \Delta$ over the larger set $\{\Delta^\top M \Delta \leq \rho^2\}$ by (c). Hence $f(\rho) = \rho \sqrt{g^\top M^+ g}$.

(d3). Let $\rho_b < \rho < \rho_{\text{sat}}$. Then $R(0) > (\rho/r)^2 > \lim_{\nu \rightarrow \infty} R(\nu)$, and R is continuous and strictly decreasing, so there is a unique root $\nu_\rho \in (0, \infty)$. Let $c = \|\Delta(\nu_\rho)\| > 0$ and $\Delta^* = r\Delta(\nu_\rho)/c$. Then $\|\Delta^*\| = r$ and $\Delta^{*\top} M \Delta^* = r^2 R(\nu_\rho) = \rho^2$, so $\Delta^* \in K(\rho)$ with both constraints active. Put $\lambda = c/(2r) > 0$ and $\mu = \nu_\rho \lambda > 0$. Then $2\lambda\Delta^* + 2\mu M \Delta^* = (c/r)(I + \nu_\rho M)\Delta^* = (I + \nu_\rho M)\Delta(\nu_\rho) = g$. The Lagrangian $L(\Delta) = g^\top \Delta - \lambda(\|\Delta\|^2 - r^2) - \mu(\Delta^\top M \Delta - \rho^2)$ is concave in Δ ($\lambda, \mu \geq 0, M \succeq 0$) with $\nabla L(\Delta^*) = g - 2\lambda\Delta^* - 2\mu M \Delta^* = 0$, so Δ^* maximizes L over $\mathbb{R}^n$. For every $\Delta \in K(\rho)$, $g^\top \Delta \leq L(\Delta) \leq L(\Delta^*) = g^\top \Delta^*$, the last equality because both constraints are active at Δ^* . Hence $f(\rho) = g^\top \Delta^* = r g^\top \Delta(\nu_\rho)/\|\Delta(\nu_\rho)\|$.

Coverage. If $g \in \text{range}(M)$, (d2), (d3), (d1) cover $[0, \rho_b]$, $(\rho_b, \rho_{\text{sat}})$, $[\rho_{\text{sat}}, \infty)$. If $g \notin \text{range}(M)$, then $\rho_b = 0$, (d3) and (d1) cover $(0, \infty)$, and $\rho = 0$ is Theorem 1(a), whose proof uses only (a) of this proposition (no circularity). $\square$

Theorem 1 gives the harm ceiling $\bar{\mathcal{F}}_W$ under output-level and state-level injection. Parts (b) and (c) are the analogues, for the excess harm of an attacker, of Paper 1, Theorem 1 (output level) and Paper 1, Theorem 3 (the finite-horizon version of its Theorem 2) and Corollary 3 (state level) [1]; the proof re-derives the linear algebra it needs and does not rely on those results. Part (d) is the analogue of the remark that follows Paper 1, Corollary 3, where “generic” means Lebesgue-almost every A for a fixed $C \neq 0$, as defined there; as Paper 1 stresses, this is a statement about a full-measure set of matrices, not a claim that the dynamics of a deployed market are generic.

Theorem 1 (Harm ceiling). *Assume Assumption 1 and Assumption 3, and write $\Pi_{\mathcal{B}_W}$ for the orthogonal projection onto the blind subspace $\mathcal{B}_W = \ker G_W$. (a) $\bar{\mathcal{F}}_W = \lim_{\varepsilon \downarrow 0} \mathcal{F}_W(\varepsilon) = \mathcal{F}_W(0) = r \|\Pi_{\mathcal{B}_W} g\|_2$, attained at $\Delta = r \Pi_{\mathcal{B}_W} g / \|\Pi_{\mathcal{B}_W} g\|_2$ when $\Pi_{\mathcal{B}_W} g \neq 0$. (b) (Output level, Paper 1 Definition D1.) $\mathcal{B}_W = \ker C$, so $\bar{\mathcal{F}}_W = r \|\Pi_{\ker C} g\|_2$. (c) (State level, Paper 1 Definition D2.) If $W \supseteq \{\tau, \dots, \tau + H - 1\}$, then $\mathcal{B}_W \subseteq \bigcap_{j=0}^{H-1} \ker(CA^j)$, with equality if W contains no time $t \geq \tau + H$. If $H \geq n$, then $\mathcal{B}_W = \mathcal{N}(A, C)$, the unobservable subspace (Paper 1, Definition D6), so $\bar{\mathcal{F}}_W = r \|\Pi_{\mathcal{N}(A, C)} g\|_2$, which is zero whenever (A, C) is observable. (d) (Genericity.) For fixed $C \neq 0$, (A, C) is observable for Lebesgue-almost every $A \in \mathbb{R}^{n \times n}$. Hence, at the state level with $H \geq n$, the harm ceiling is zero excess harm for Lebesgue-almost every A .*

Proof. (a). By Proposition 2(a), f is right-continuous at 0 and nondecreasing, and $\rho_\varepsilon \downarrow 0$ as $\varepsilon \downarrow 0$, so $\lim_{\varepsilon \downarrow 0} \mathcal{F}_W(\varepsilon) = f(0) = \mathcal{F}_W(0)$. Now $K(0) = \{\Delta : \|\Delta\|_2 \leq r, \Delta^\top M_W \Delta = 0\} = \{\Delta \in \mathcal{B}_W : \|\Delta\|_2 \leq r\}$, because $\Delta^\top M_W \Delta = \|V_W^{-1/2} G_W \Delta\|_2^2$ vanishes iff $G_W \Delta = 0$ ($V_W \succ 0$). For $\Delta \in \mathcal{B}_W$, $g^\top \Delta = (\Pi_{\mathcal{B}_W} g)^\top \Delta \leq r \|\Pi_{\mathcal{B}_W} g\|_2$ by Cauchy-Schwarz, with equality at the stated point (which lies in $\mathcal{B}_W$, with norm r); if $\Pi_{\mathcal{B}_W} g = 0$ the value 0 is attained at $\Delta = 0$.

(b). At the output level the block row of G_W at $t \in W$ is 0 for $t < \tau$ and C for $t \geq \tau$ (Definition 2), and W contains τ . So $G_W \Delta = 0$ iff $C \Delta = 0$.

(c). At the state level the block row at $t = \tau + j$ is CS_j , $S_j = \sum_{i=0}^j A^i$, and 0 for $t < \tau$. Hence $\mathcal{B}_W = \bigcap_{t \in W, t \geq \tau} \ker(CS_{t-\tau}) \subseteq \bigcap_{j=0}^{H-1} \ker(CS_j)$, with equality if W has no time $t \geq \tau + H$. Since $CS_0 = C$ and $CA^j = CS_j - CS_{j-1}$ for $j \geq 1$, while $CS_j = \sum_{i \leq j} CA^i$, a vector is in $\ker(CS_j)$ for all $j < H$ iff it is in $\ker(CA^j)$ for all $j < H$ (induction on j). This gives the inclusion and the equality case. Let $H \geq n$. Then $\mathcal{B}_W \subseteq \bigcap_{j=0}^{n-1} \ker(CA^j) = \mathcal{N}(A, C)$. Conversely, by Cayley-Hamilton every A^k , $k \geq n$, is a linear combination of $I, A, \dots, A^{n-1}$, so $\Delta \in \mathcal{N}(A, C)$ gives $CA^k \Delta = 0$ for all $k \geq 0$, hence $CS_j \Delta = 0$ for all $j \geq 0$, hence $\Delta \in \mathcal{B}_W$ whatever the other times in W . So $\mathcal{B}_W = \mathcal{N}(A, C)$, and (a) gives the ceiling; if (A, C) is observable, $\mathcal{N}(A, C) = \{0\}$ and the ceiling is 0.

(d). (A, C) is observable iff the observability matrix $\mathcal{O}(A) = [C; CA; \dots; CA^{n-1}]$ has rank n , iff $P_C(A) \neq 0$, where $P_C(A)$ is the sum of the squares of all $n \times n$ minors of $\mathcal{O}(A)$, a polynomial in the entries of A . P_C is not the zero polynomial: let $c^\top \neq 0$ be a row of C , let $N = \sum_{i=1}^{n-1} e_i e_{i+1}^\top$ (so $e_1^\top N^k = e_{k+1}^\top$ for $k < n$), and let T be invertible with $T^\top c = e_1$ (it exists since $c \neq 0$). For $A_0 = TNT^{-1}$, $c^\top A_0^k = e_1^\top N^k T^{-1} = e_{k+1}^\top T^{-1}$, $k = 0, \dots, n-1$: these are the n rows of T^{-1} , linearly independent, and they are rows of $\mathcal{O}(A_0)$. So $P_C(A_0) \neq 0$. The zero set of a non-zero polynomial on $\mathbb{R}^{n \times n}$ is Lebesgue-null (standard; by induction on the number of variables with Fubini). Hence (A, C) is observable for Lebesgue-almost every A , and by (c) the ceiling is zero for those A when $H \geq n$. $\square$

Exact stealth ($\varepsilon = 0$) is therefore always available, since the attacker can reproduce the fault’s injection; under state feedback on a window of at least n steps and for generic dynamics it is the only exactly stealthy attack, so it yields zero excess harm, and any positive excess harm costs a strictly positive ε .

B. Exploitable commitments

Proposition 3 shows that a threshold rule committed to a nominal attack model detects a payload only through one scalar, so that an attacker who knows the rule can choose a payload the rule detects only at its false-alarm rate. Which rule the operator should commit to instead is the question of the numerical study of Section V.

Proposition 3 (The committed threshold rule is blind off its nominal signature). *Assume Assumption 1. Fix a nominal model $\bar{x}$, a threshold $q^\dagger \in (0, 1)$ ($q^\dagger = q^*$ for $\pi_{\bar{x}}^{\text{thr}}$) and the rule π that contains iff $\bar{q}(Y_W) \geq q^\dagger$. Put $w = M_W(\bar{x} - x_F)$, $\bar{D} = d_W(\bar{x} - x_F)$, $\xi^\dagger = \log((1-p)q^\dagger/(p(1-q^\dagger)))$, and $\Delta = x_K - x_F$. (a) If $\bar{D} > 0$, then for every $x_K \in \mathbb{R}^n$*

$$\beta_\pi(x_K) = \Phi\left(\frac{w^\top \Delta - \xi^\dagger}{\bar{D}} - \frac{\bar{D}}{2}\right), \quad \alpha_\pi = \Phi\left(-\frac{\xi^\dagger}{\bar{D}} - \frac{\bar{D}}{2}\right).$$

So β_π depends on x_K only through the scalar $w^\top \Delta = (G_W \bar{x} - G_W x_F)^\top V_W^{-1} G_W \Delta$ and is strictly increasing in it: $\beta_\pi(x_K) = \alpha_\pi$ iff $w^\top \Delta = 0$, and $\beta_\pi(x_K) < \alpha_\pi$ iff $w^\top \Delta < 0$. (b) If $\bar{D} = 0$, then $\bar{q}(Y_W) = p$ almost

surely, π is constant, and $\beta_\pi(x_K) = \alpha_\pi$ for every x_K . (c) Under Assumption 3, the detection-harm frontier at the false-alarm rate is $\Psi_\pi(\alpha_\pi) = r\|g\|_2$ if $w^\top g \leq 0$, and $\Psi_\pi(\alpha_\pi) = r\|\Pi_{w^\perp} g\|_2$ if $w^\top g > 0$, where $\Pi_{w^\perp}$ is the orthogonal projection onto $\{u : w^\top u = 0\}$; in the second case the supremum is attained by a payload with $\beta_\pi = \alpha_\pi$. (d) Under Assumption 3, let $\lambda_\perp^2 = \max\{u^\top M_W u : \|u\|_2 = 1, w^\top u = 0\}$. The largest total variation among payloads the rule does not see is $\max\{\text{TV}(P_x^W, P_{x_F}^W) : x \in \mathcal{X}_r, \beta_\pi(x) = \alpha_\pi\} = 2\Phi(r\lambda_\perp/2) - 1$, and it is strictly positive whenever $\text{rank } M_W \geq 2$.

Proof. Write $G = G_W$, $V = V_W$, $M = M_W$, $\mu = \mu_W$, and $\bar{s} = G(\bar{x} - x_F)$, so $\bar{D}^2 = \bar{s}^\top V^{-1} \bar{s}$ and $w = G^\top V^{-1} \bar{s}$.

The statistic. The densities of $P_{\bar{x}}^W$ and $P_{x_F}^W$ are Gaussian with common covariance V (Definition 2), so

$$\log \frac{f_{\bar{x}}^W(y)}{f_{x_F}^W(y)} = \bar{s}^\top V^{-1}(y - \mu - Gx_F) - \frac{1}{2}\bar{D}^2.$$

For $q^\dagger \in (0, 1)$, $\bar{q}(y) \geq q^\dagger$ iff $\log(pf_{\bar{x}}^W(y)/(1-p)f_{x_F}^W(y)) \geq \log(q^\dagger/(1-q^\dagger))$ iff $T(y) := \bar{s}^\top V^{-1}(y - \mu - Gx_F) \geq \xi^\dagger + \frac{1}{2}\bar{D}^2$.

(a). Under $P_{x_K}^W$, $Y_W - \mu - Gx_F \sim \mathcal{N}(G\Delta, V)$, so $T(Y_W) \sim \mathcal{N}(\bar{s}^\top V^{-1}G\Delta, \bar{s}^\top V^{-1}VV^{-1}\bar{s}) = \mathcal{N}(w^\top \Delta, \bar{D}^2)$. If $\bar{D} > 0$ this law has no atom, so

$$\beta_\pi(x_K) = P(T \geq \xi^\dagger + \frac{1}{2}\bar{D}^2) = 1 - \Phi\left(\frac{\xi^\dagger + \bar{D}^2/2 - w^\top \Delta}{\bar{D}}\right) = \Phi\left(\frac{w^\top \Delta - \xi^\dagger}{\bar{D}}\right).$$

The false-alarm rate is the case $x_K = x_F$, $\Delta = 0$. Also $w^\top \Delta = \bar{s}^\top V^{-1}G\Delta = (G\bar{x} - Gx_F)^\top V^{-1}G\Delta$ and $w = G^\top V^{-1}G(\bar{x} - x_F) = M(\bar{x} - x_F)$. Since Φ is strictly increasing, $\beta_\pi(x_K)$ is a strictly increasing function of $w^\top \Delta$, equal to α_π exactly at $w^\top \Delta = 0$.

(b). If $\bar{D} = 0$, then $\bar{s} = 0$ (as $V^{-1} \succ 0$), so $P_{\bar{x}}^W = P_{x_F}^W$, $f_{\bar{x}}^W = f_{x_F}^W$ and $\bar{q} \equiv p$ (almost surely, where the densities are positive). The rule contains always (if $p \geq q^\dagger$) or never, so $\beta_\pi(x_K) = \alpha_\pi \in \{0, 1\}$ for every x_K .

(c). Note $w = 0$ iff $\bar{D} = 0$, since $\bar{D}^2 = (\bar{x} - x_F)^\top M(\bar{x} - x_F)$ and $M \succeq 0$. If $w = 0$, every payload has $\beta_\pi = \alpha_\pi$ by (b), so $\Psi_\pi(\alpha_\pi) = \max_{\|\Delta\| \leq r} g^\top \Delta = r\|g\|$, consistent with the case $w^\top g \leq 0$. If $w \neq 0$, by (a) $\beta_\pi(x) \leq \alpha_\pi$ iff $w^\top \Delta \leq 0$, so $\Psi_\pi(\alpha_\pi) = \sup\{g^\top \Delta : \|\Delta\| \leq r, w^\top \Delta \leq 0\}$. If $w^\top g \leq 0$, $\Delta = rg/\|g\|$ (or 0 if $g = 0$) is feasible and maximizes $g^\top \Delta$ over the whole ball: the value is $r\|g\|$. If $w^\top g > 0$, write $g = \Pi_{w^\perp} g + \frac{w^\top g}{\|w\|^2} w$. For feasible Δ , $g^\top \Delta = (\Pi_{w^\perp} g)^\top \Delta + \frac{w^\top g}{\|w\|^2} w^\top \Delta \leq (\Pi_{w^\perp} g)^\top \Delta \leq r\|\Pi_{w^\perp} g\|$. The bound is attained at $\Delta = r\Pi_{w^\perp} g/\|\Pi_{w^\perp} g\|$ (or 0 if $\Pi_{w^\perp} g = 0$), which has $w^\top \Delta = 0$, hence $\beta_\pi = \alpha_\pi$.

(d). By (a) and (b), $\{x \in \mathcal{X}_r : \beta_\pi(x) = \alpha_\pi\} = \{x_F + \Delta : \|\Delta\| \leq r, w^\top \Delta = 0\}$ (for $w = 0$, the whole ball). By Definition 4, $\text{TV} = 2\Phi(d_W(\Delta)/2) - 1$, strictly increasing in $d_W(\Delta) = \sqrt{\Delta^\top M \Delta}$. The maximum of $\Delta^\top M \Delta$ over the ball intersected with the subspace $w^\perp$ is $r^2 \lambda_\perp^2$ (a quadratic form over a ball in a subspace is maximized on its sphere, at a top eigenvector of the compression of M to the subspace). Hence the maximum TV is $2\Phi(r\lambda_\perp/2) - 1$. It is 0 iff $\lambda_\perp = 0$, i.e. iff $u^\top M u = 0$, hence $M u = 0$ ($M \succeq 0$), for every $u \in w^\perp$. Then $\ker M$ contains a

subspace of dimension at least $n - 1$, so $\text{rank } M \leq 1$. So the maximum is strictly positive when $\text{rank } M \geq 2$. $\square$

C. The verification margin

Proposition 4 bounds the detection probability of every ε -stealthy attacker against a committed two-threshold rule with verification. The rule is the one proposed in Paper 2 [2], whose optimality result is not yet independently verified; the inequality itself uses only the ordering of the two thresholds and Assumption 6.

Proposition 4 (Verification margin). *Assume Assumption 1 and Assumption 6, with $s_v, t_v \in [0, 1]$. Let $\pi = \pi_{\bar{x}}^{\text{ver}}$ with thresholds $q_1 \leq q_2$ (Paper 2's, or any others), $\alpha = \alpha_\pi$ and m_F the fault verification mass (Definition 6). For every $\varepsilon \in [0, 1]$ and every x_K with $\text{TV}(P_{x_K}^W, P_{x_F}^W) \leq \varepsilon$ (in particular every $x_K \in \mathcal{E}_W(\varepsilon)$), with $\Delta = x_K - x_F$: (a) $(s_v + t_v - 1)m_F - \varepsilon \leq \beta_\pi(x_K) - \alpha \leq (s_v + t_v - 1)m_F + \varepsilon$; (b) $1 - \alpha - (s_v + t_v - 1)m_F + \varepsilon \geq 0$ and $U_A(\pi, x_K) \leq h(\Delta)^+ (1 - \alpha - (s_v + t_v - 1)m_F + \varepsilon)$, where $h^+ = \max\{h, 0\}$; in particular, if $h(\Delta) \geq 0$, $U_A(\pi, x_K) \leq h(\Delta)(1 - \alpha - (s_v + t_v - 1)m_F + \varepsilon)$. Under Assumption 5 ($s_v + t_v > 1$), the margin $(s_v + t_v - 1)m_F$ is positive iff $m_F > 0$; if the verification region is empty ($q_1 = q_2$), $m_F = 0$ and (a) reduces to Proposition 1(b).*

Proof. Write $\bar{q} = \frac{q_1 + q_2}{2}$. By Definition 6,

$$\phi^K - \phi^F = (s_v - (1 - t_v))\chi = (s_v + t_v - 1)\chi.$$

By Proposition 1(c) (which uses Assumption 6), $\beta_\pi(x_K) = \mathbb{E}_{x_K}[\phi^K]$ and $\alpha = \mathbb{E}_{x_F}[\phi^F]$, and $\mathbb{E}_{x_F}[\chi] = m_F$ by definition. Hence

$$\mathbb{E}_{x_F}[\phi^K] = \mathbb{E}_{x_F}[\phi^F] + (s_v + t_v - 1)\mathbb{E}_{x_F}[\chi] = \alpha + (s_v + t_v - 1)m_F.$$

(a). By Proposition 1(a) with $\phi = \phi^K$ (a measurable map into $[0, 1]$), $|\beta_\pi(x_K) - \mathbb{E}_{x_F}[\phi^K]| \leq \varepsilon$. Substituting the identity above gives both inequalities.

(b). Let $B = 1 - \alpha - (s_v + t_v - 1)m_F + \varepsilon$. By (a), $\beta_\pi(x_K) \geq 1 - B$, so $B \geq 1 - \beta_\pi(x_K) \geq 0$. Since $c_A \geq 0$ and $\beta_\pi \geq 0$, $U_A(\pi, x_K) = h(\Delta)(1 - \beta_\pi) - c_A \beta_\pi \leq h(\Delta)(1 - \beta_\pi)$. If $h(\Delta) \geq 0$, then $h(\Delta)(1 - \beta_\pi) \leq h(\Delta)B$. If $h(\Delta) < 0$, then $h(\Delta)(1 - \beta_\pi) \leq 0 = h(\Delta)^+ B$. In both cases $U_A \leq h(\Delta)^+ B$.

Last sentence. If $s_v + t_v > 1$, $(s_v + t_v - 1)m_F > 0$ iff $m_F > 0$. If $q_1 = q_2$, the interval (q_1, q_2) is empty, $\chi \equiv 0$, $m_F = 0$ and $\phi^K = \phi^F$, so the rule is a rule without verification and (a) is Proposition 1(b). $\square$

V. NUMERICAL ILLUSTRATION

This section will illustrate the results of Section IV on a small linear-Gaussian instance: the stealth-harm frontier as a function of ε under output-level and state-level injection, and, for rules with verification, the operator's optimal commitment computed numerically, to compare the Stackelberg-optimal verification region with the non-strategic one of Paper 2 [2].

VI. DISCUSSION

The results are stated for an operator who commits. Under simultaneous play the operator's rule is a best response to the attacker's possibly mixed strategy, and its posterior is computed under the attacker's actual law rather than a nominal one. Simultaneous-play formulations of detection against stealthy attackers include the Bayesian Nash equilibrium of [16] and the zero-sum saddle point of [19]. The discussion separates the results that bound every pair of strategies, and therefore hold under simultaneous play as well, from those that depend on commitment. It also sets out the limits of the model: an attacker who also compromises the authorization channel, for instance with stolen credentials, violates the separate-channel assumption and is out of scope; under an absolute-harm payoff the ceiling is the fault's own harm rather than zero; and the attacker is assumed to know the fault instance it imitates.

VII. CONCLUSION

This paper extends the attack-fault ambiguity framework to an attacker who knows the operator's rule. It asks what stealth costs such an attacker in harm, what harm remains when stealth is exact, whether a rule committed to a nominal attack model can be exploited, and how much a verification step on a separate channel protects the operator.

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